

README TableTask Movesense ECG/HR Analysis

Veronica / Cirkeline TableTask dyadic physiology pipeline

Concise technical guide for GitHub users and future MC Lab research assistants

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1. Purpose of this repository

This repository documents the TableTask Movesense ECG/HR analysis pipeline used to process dyadic Movesense recordings, segment them into task windows, assess signal quality/reliability, and compute Lena-style dyadic synchrony metrics.

- The pipeline starts from raw Movesense ECG recordings and task timing/metadata files.
- It produces corrected beat-to-beat RR intervals, corrected 4 Hz heart-rate time series, trial-level paired HR segments, RR reliability labels, and trial-level dyadic synchrony metrics.
- The final Step 4 analysis is based on Lena Adel's tutorial dyadic Movesense pipeline, especially Notebook 04 and Notebook 04b, with Veronica-specific adaptations for TableTask trial segmentation and QC.
- Rows are not silently removed in the final tables. Use the explicit QC and main-analysis columns to decide what to include in downstream statistical models.

2. Expected top-level folder structure

Run scripts from the top-level TableTask folder unless a script header says otherwise. The current GitHub/work folder is organized as follows:

```
TableTask/
  processed_ecg_hr/
  qc_outputs/
  Table MoveSense Sensor/
  aggregated_data.csv
  table_timestamps.csv
  Table Task 2026.xlsx
  README_TableTask_QC_CommonSensorsQuestion_Answers_clean.docx
  Paper_OddEvenReliability.pdf
  01_script_prepare_tabletask_metadata_qc_EG_final.py
  02_script_process_ecg_to_hr_EG_final.py
  02a_script_organize_step2_plots_by_sensor_EG.py
  02b_script_extract_pretrial_2min_qc_EG_v2_final.py
  02c_script_organize_step2b_pretrial_qc_plots_by_sensor_EG_final.py
  03a_script_rr_odd_even_reliability_QC_EG.py
  04_script_trial_synchrony_lena_notebooks4_4b_dual_lag_labels_v14_EG.py
```

3. Core input files and folders

Item	Role in pipeline
Table MoveSense Sensor/	Raw Movesense session folders. Each complete dyad session should contain two sensor recordings. Needed for ECG processing in Step 2, not needed by Step 4.
aggregated_data.csv	Behavioral/trial-level task data used to attach condition and task covariates to analysis windows.

table_timestamps.csv	Timing file used to define task windows and align task timing to physiological recordings.
Table Task 2026.xlsx	Project metadata/QC workbook used by Step 1 to construct analysis-unit metadata and exclusion flags.
qc_outputs/veronica_analysis_units.csv	Step 1 master analysis-unit file: one row per expected candidate window/session.
processed_ecg_hr/hr_interpolated_4hz/	Step 2 corrected 4 Hz HR time series for each sensor recording; used by Step 3.
processed_ecg_hr/rpeak_rr_corrected/	Step 2 corrected beat-to-beat RR files; used by Step 4 for Lena 04b Poincare/Bland-Altman metrics.
processed_ecg_hr/trial_segments_4hz/paired_trial_hr/	Step 3 paired dyad-by-trial HR files on a common 4 Hz grid; direct inputs to Step 4.

4. Analysis-unit design

Each complete dyad/session is expected to have nine candidate windows:

Window	Trial label	Role
cw01	trial00	Practice window. Computed and plotted for QC/diagnostics, but not part of the primary FF vs FE analysis.
cw02-cw09	trial01-trial08	Eight real task trials. These are the intended FF/FE experimental trials.

- Current dataset: Step 1 created 162 analysis units (18 sessions x 9 candidate windows).
- Current dataset: Step 3/Step 4 processed 156 paired HR windows, so six expected windows did not become paired HR files upstream of Step 4.
- If a dyad has fewer than nine final plots, check Step 1 and Step 3 summaries before assuming a Step 4 problem.
- Primary analyses should use `main_analysis_eligible == True`, then apply approved QC/reliability filters.

5. Script-by-script workflow

Step/script	Purpose	Main inputs	Main outputs/checks
01 metadata/QC	Builds project-specific metadata and analysis-unit table: dyad/session windows, trial labels, conditions, exclusion flags, and timing fields.	Table Task 2026.xlsx; aggregated_data.csv; table_timestamps.csv; metadata/QC information.	qc_outputs/veronica_analysis_units.csv/.xlsx; TableTask_Master_QC.xlsx; metadata_qc_summary.txt. Current expectation: 162 analysis-unit rows.
02 ECG -> HR	Processes raw Movesense ECG for each sensor recording. Adapted from Lena Notebook 03: ECG cleaning, R-peak detection/correction, RR intervals, corrected HR, HRV/QC plots.	Table MoveSense Sensor/ and Step 1 metadata.	rpeak_rr_corrected/; hr_interpolated_4hz/; hrv_metrics/; Step 2 QC plots and combined PDFs. Current expectation: 36 sensor recordings.

02a plot organizer	Organizes Step 2 plots by sensor/session for easier review. Does not change analysis values.	processed_ecg_hr/plots/ and Step 2 outputs.	Organized plot folders under processed_ecg_hr/plots/.
02b pretrial QC	Extracts/summarizes the first 2-minute pretrial ECG/HR period for each sensor.	Step 2 corrected HR/RR outputs.	processed_ecg_hr/pretrial_2min_qc_v2/ with pretrial summaries and plots.
02c pretrial plot organizer	Organizes Step 2b pretrial QC plots. Does not change analysis values.	pretrial_2min_qc_v2/ outputs.	Organized Step 2b plot folders.
03 trial segmentation	Cuts full-session corrected 4 Hz HR into TableTask windows and creates paired dyad trial files on a common 4 Hz time grid. Veronica-specific bridge between Lena-style ECG processing and trial-level synchrony.	hr_interpolated_4hz/; qc_outputs/veronica_analysis_units.csv.	trial_segments_4hz/paired_trial_hr/*.csv; qc/03_trial_segmentation_summary.csv. Current expectation: 156 paired rows.
03a RR reliability	Computes odd-even RR reliability and dyad-level reliability labels. QC guidance, not automatic final exclusion unless approved.	Corrected RR files; Step 3 trial metadata.	rr_reliability_qc/qc/3a_trial_dyad_rr_reliability.csv/.xlsx. Labels: reliable, caution, low, unavailable.
04 v14 synchrony	Computes trial-level dyadic synchrony and Lena-style plots. Adapted from Lena Notebook 04 and 04b with Veronica-specific metadata and strict direct-RR extraction.	Step 3 paired HR; Step 3 summary; Step 3a reliability; Step 2 corrected RR.	trial_synchrony_lena_step4_EG/tables/ and four plot folders. Current run: 156/156 successful, 0 failures, 115 main-analysis eligible.

6. Step details

6.1 Step 1: metadata, timing, and QC table

Step 1 is project-specific and is not directly from Lena's tutorial. It creates the metadata backbone used by all later scripts. The most important output is *qc_outputs/veronica_analysis_units.csv*, which defines each expected dyad/session candidate window and carries timing, condition, and QC/exclusion fields.

- Check that each complete session has cw01-cw09 unless there is a known metadata or QC reason.
- Use `exclude_from_main_analysis` and `exclude_reason` to understand which rows should not enter the primary FF/FE analysis.
- Practice rows are useful for diagnostics but should not be treated as experimental FF/FE trials.

6.2 Step 2: raw ECG processing and HR extraction

Step 2 is the main ECG-processing step and is adapted from Lena Notebook 03. It processes raw ECG at the sensor-recording level, not the trial level.

- Cleans raw ECG using NeuroKit-style ECG preprocessing.
- Detects and corrects R peaks.
- Computes beat-to-beat RR intervals and corrected HR.
- Interpolates corrected HR to a 4 Hz time series for later trial segmentation.
- Saves corrected RR files used later by Step 4 Poincare/Bland-Altman calculations.
- Saves diagnostic plots: raw vs clean ECG, R peaks, HR before/after correction, HR time series, HRV PSD, Poincare, and dyad-session previews.

6.3 Step 2b: pretrial 2-minute QC

Step 2b extracts a standardized pretrial window for each sensor. This provides a sensor-level QC view separate from task trials and helps carry pretrial QC flags into later summaries.

6.4 Step 3: trial-level HR segmentation

Step 3 is a Veronica-specific segmentation bridge. It takes corrected 4 Hz HR files from Step 2, cuts them into TableTask windows defined by Step 1, and writes paired dyad files on a common trial-relative time grid.

- Step 4 can only process windows that exist in `processed_ecg_hr/trial_segments_4hz/paired_trial_hr/`.
- If a dyad has fewer than nine final plots, first check `processed_ecg_hr/trial_segments_4hz/qc/03_trial_segmentation_summary.csv`.
- Current dataset: Step 3 generated 156 paired HR files from 162 expected analysis units.

6.5 Step 3a: RR odd-even reliability QC

Step 3a estimates within-recording/trial reliability using odd-even RR logic. It provides reliability labels that help identify unstable or low-quality physiological segments.

Reliability label	Current threshold used
reliable	$r \geq 0.75$
caution	$0.50 \leq r < 0.75$
low	$r < 0.50$
unavailable	Too few valid RR intervals or undefined reliability

Important: These thresholds are QC guidance. They should not become final exclusion rules without Veronica's approval.

6.6 Step 4 v14: dyadic synchrony and Lena-style outputs

Step 4 v14 is the final analysis stage in this repository. It is adapted from Lena Notebook 04 and Notebook 04b, but applied to Veronica's paired trial-level TableTask files. It does not rerun ECG processing and does not require the raw Movesense folder.

- Notebook 04-style synchrony: paired finite HR samples, linear detrending, z-scoring, then Pearson correlation, windowed Pearson, PLI/PLV, cross-correlation, coherence, envelope correlation, and optional CRQA.
- Notebook 04b-style Poincare/Bland-Altman: uses Step 2 corrected beat-to-beat RR files, not RR reconstructed from 4 Hz HR, when `--require-direct-rr-for-poincare` is used.

- Notebook 04b-style leader-follower: cross-correlation lag, lag-1 directed correlations, phase directionality, overall leader classification, and optional Granger causality.
- Cross-correlation lag is saved with both Lena-convention and SciPy-convention direction labels. Lena's Notebook 04b interprets positive lag as participant A leading participant B, whereas the common SciPy signal-processing interpretation of `correlate(A, B)` can imply the opposite direction. Use numeric `lag_s` columns as the primary directionality result and interpret labels carefully.

7. Expected Step 4 v14 outputs

Output	Purpose
tables/04_step4_trial_level_synchrony_metrics_EG.csv/.xlsx	Primary trial-level table. One row per successful paired dyad-window. Use this table for downstream analysis.
tables/04_step4_window_level_synchrony_metrics_EG.csv/.xlsx	Window-level synchrony table. Multiple rows per trial if enough data exist for multiple sliding windows.
tables/04_step4_condition_level_summary_all_windows_EG.csv/.xlsx	Condition-level descriptive summary across all computed windows/trials.
tables/04_step4_condition_level_summary_main_analysis_EG.csv/.xlsx	Condition-level descriptive summary restricted to rows marked <code>main_analysis_eligible</code> .
tables/04_step4_file_manifest_EG.csv	List of Step 3 paired HR files found by Step 4. Useful for confirming what Step 4 could process.
tables/04_step4_plot_manifest_EG.csv	One row per attempted plot. Use this to check saved/skipped/failed plot status. v14 attempts four plot types per processed paired HR file.
tables/04_step4_failures_EG.csv	Should be empty in a clean run. If not empty, inspect the error column.
tables/04_step4_run_summary_EG.txt/.json	Run-level summary: script version, input/output paths, row counts, failure counts, and settings.
README_04_step4_outputs_EG.txt	Short automatic output README written by the Step 4 script.

8. Expected Step 4 v14 plot folders

Folder	What it shows	Expected current count
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lena_04_windowed_cardiac_synchrony/	Two-panel trial plot: top = windowed Pearson HR synchrony; bottom = windowed PLI. Main single-trial synchrony diagnostic plot.	156 PNGs if all paired files plot successfully.
lena_04b_comprehensive_poincare/	Multi-panel Poincare/Bland-Altman view using direct corrected beat-to-beat RR intervals.	156 PNGs if all paired files plot successfully.
lena_04b_overlaid_ellipses/	Participant A and B Poincare scatter/ellipse overlay for each dyad-window.	156 PNGs if all paired files plot successfully.
lena_04b_temporal_poincare_patterns/	Temporal Poincare pattern panels across windowed segments within a trial.	156 PNGs if all paired files plot successfully.

9. How to interpret and filter final outputs

- The primary analysis table is 04_step4_trial_level_synchrony_metrics_EG.csv.
- Use `main_analysis_eligible == True` for the primary FF/FE dataset. Practice and excluded rows are kept for transparency and QC.
- For stricter descriptive summaries, consider filtering `trial_segment_qc_flag == "clean"` and `dyad_rr_reliability_label == "reliable"`. Treat this as a QC sensitivity filter unless Veronica has approved it as the main analysis rule.
- Do not interpret a single trial plot as a condition effect. Single-trial plots are diagnostic. Condition effects require analysis across dyads/trials, ideally with an appropriate mixed-effects/statistical model.
- A trial with one window shows one dot because the 30-second windowing produced only one valid window. This does not mean there was one HR sample; it means one window-level synchrony estimate was computed.
- If a dyad has fewer than nine windows, check Step 1 `veronica_analysis_units.csv` and Step 3 `03_trial_segmentation_summary.csv`. Step 4 cannot create windows that were not produced by Step 3.

10. Key Step 4 metrics glossary

Metric/column family	Meaning
pearson_r / spearman_r	Trial-level association between participants' HR signals after Lena-style preprocessing. Pearson is linear; Spearman is monotonic.
windowed_pearson_mean	Average of sliding-window Pearson synchrony values within a trial.
phase_locking_index_pli / phase_lag_index	Phase-lag consistency between HR signals using Hilbert phase. Values closer to 1 indicate more consistent non-zero phase lag.
phase_locking_value_plv	Phase-locking value. Supplementary phase metric; values closer to 1 indicate stronger phase consistency.
crosscorr_* and lf_xcorr_*	Lagged association and leader-follower indicators. Interpret numeric lag carefully and report the convention used.
mean_coherence_0p04_0p40_hz	Mean coherence in the 0.04-0.40 Hz band, when enough samples exist. Short trials may have NaN by design.
envelope_corr	Correlation between Hilbert amplitude envelopes of the two HR signals.

A_rr_* / B_rr_* / dyad_mean_rr_*	Poincare-style RR variability metrics computed from corrected beat-to-beat RR intervals.
granger_*	Optional Granger-causality results from statsmodels. Exploratory and sensitive to sample length and assumptions.
crqa_*	Optional cross-recurrence quantification analysis outputs if PyRQA is installed; otherwise unavailable/NaN.

End of README.